ASSIGNMENT #2

33. PSO – CRM System

Software Scope and Effort Estimation

Pakistan State Oil Company Limited
Customer Relationship Management System

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TASK (b) – COCOMO-II Early Design Effort Estimation

This section applies the COCOMO-II Early Design model to estimate the software development effort for the PSO CRM system. All twelve parameters are evaluated and justified with specific reference to the project context. The model is applied in five sequential steps as prescribed by Boehm et al. (2000).

Step 1: SLOC Estimation from UFP

The starting point for COCOMO-II Early Design is a Source Lines of Code (SLOC) estimate derived from the Unadjusted Function Points computed in Task (a).

- Total UFP established in Task (a): 45 Function Points
- Technology selection: The PSO CRM system employs a mixed-technology stack consisting of Java/Spring Boot for backend microservices, C/C++ for the legacy POS sync agent, and C# / .NET MAUI for the fleet management portal. The industry-standard conversion factor for this mixed stack is 50 LOC per Function Point.
- SLOC Calculation: 45 FP × 50 LOC/FP = 2,250 SLOC
- KLOC Conversion: 2,250 SLOC ÷ 1,000 = 2.25 KLOC

Justification for 50 LOC/FP: The ISBSG (International Software Benchmarking Standards Group) reports average conversion rates of 46–55 LOC/FP for Java and C# enterprise applications. Given the inclusion of a legacy C/C++ component (the POS sync agent), which tends to require more verbose code, the midpoint value of 50 LOC/FP is the most appropriate and defensible selection for this project.

Step 2: Scale Factors (SF) — Exponent B Calculation

Scale Factors capture the non-linear relationship between project size and effort. The five COCOMO-II scale factors below are each rated and justified in the context of the PSO CRM project. The formula for exponent B is:

$$B = 0.91 + 0.01 \times (PREC + FLEX + RESL + TEAM + PMAT)$$

Scale Factor	Rating	Value	Justification & Context
PREC (Precedentedness)	Nominal	**3.72**	PSO operates large-scale logistics and ERP systems; however, deploying a customer-facing, offline-first CRM that directly manages financial loyalty points (DigiCash) is a relatively novel undertaking for the organisation. The precedent is moderate — the technology is known but the exact domain combination is new.
FLEX (Development Flexibility)	Low	**4.05**	Extremely constrained. The system must adhere rigidly to SBP data privacy regulations, SECP compliance directives, and the predefined data exchange formats mandated by PSO's legacy SAP financial ledger. Deviations in data structure are not permissible.
RESL (Risk Resolution)	High	**2.83**	A comprehensive Risk Register was developed in Assignment 1, covering five critical risks including data integration complexity and connectivity issues. The offline-first architecture directly and proactively mitigates the highest-probability risk, yielding high risk resolution confidence.
TEAM (Team Cohesion)	Nominal	**3.29**	The development team comprises a heterogeneous group: legacy SAP integration specialists, mobile application developers, field operations managers, and new software engineers. This mix introduces moderate coordination overhead but benefits from diverse domain knowledge.
PMAT (Process Maturity)	Nominal	**4.68**	PSO's IT Department operates under standard enterprise PMO governance with defined SDLC stages, change management boards, and ISO 27001 compliance audits. Process maturity is at CMM Level 2–3, consistent with a Nominal rating.
SF Total	—	18.57	**Sum of all Scale Factor values used in exponent B calculation**

$$B = 0.91 + 0.01 \times (3.72 + 4.05 + 2.83 + 3.29 + 4.68) = 0.91 + 0.01 \times 18.57 = 0.91 + 0.1857 \approx 1.0957$$

Step 3: Effort Multipliers (EM) — Effort Adjustment Factor (EAF)

The seven COCOMO-II Early Design Effort Multipliers collectively capture project-level and environmental characteristics that scale the base effort estimate. Each multiplier is rated and justified below.

Effort Multiplier	Rating	Value	Justification & Context
RCPX (Product Reliability & Complexity)	High	**1.33**	The system manages real financial liabilities (DigiCash PKR values) and executes complex multi-node offline queuing (FR02) across 3,500+ distributed POS terminals. Data integrity failures could trigger financial discrepancies and regulatory penalties, warranting a High complexity rating.
RUSE (Developed for Reusability)	Nominal	**1.00**	The system is developed specifically for PSO's proprietary POS hardware ecosystem and SAP configuration. It is not

			architected for external resale or reuse in other fuel retail organisations, justifying a Nominal rating.
PDIF (Platform Difficulty)	High	**1.29**	The platform presents significant constraints: integration with legacy Pentium-era POS hardware (NFR05), unreliable 2G/3G connectivity at remote stations (NFR10), and a bridged SAP ERP interface that imposes strict data format requirements (FR09).
PERS (Personnel Capability)	Nominal	**1.00**	The development team consists of competent enterprise software engineers with standard capability ratings for both backend and integration domains. No exceptional capability surplus or deficit is anticipated.
PREL (Personnel Continuity)	Nominal	**1.00**	Standard corporate HR retention policies apply. Given the 12-month development timeline, moderate personnel stability is expected. No critical specialist dependencies that could cause schedule disruption have been identified.
FCIL (Facilities)	Nominal	**1.00**	PSO's IT infrastructure provides adequate development environments including servers at PSO House Karachi, standard software development tooling (IDEs, CI/CD pipelines), and VPN access for remote developers.
SCED (Required Schedule)	Nominal	**1.00**	The 12-month development and rollout schedule, as defined in the Assignment 1 business case, aligns with COCOMO II norms for a project of this size and complexity. No compression or extension is required at this stage.
EAF (Product)	**—**	**1.7157**	**1.33 × 1.00 × 1.29 × 1.00 × 1.00 × 1.00 × 1.00 = 1.7157**

$$EAF = 1.33 \times 1.00 \times 1.29 \times 1.00 \times 1.00 \times 1.00 \times 1.00 = 1.7157$$

Step 4: Final Effort Estimation

With all parameters computed, the COCOMO-II Early Design effort formula is applied as follows:

$$\text{Effort} = A \times (\text{KLOC})^B \times \text{EAF}$$

Where A = 2.94 (standard COCOMO II Early Design constant)

$$(2.25)^{1.0957} = 2.44$$

$$\text{Effort} = 2.94 \times 2.44 \times 1.7157 \approx 12.30 \text{ Person-Months}$$

Step 5: Development Duration and Team Composition

The derived effort of 12.30 Person-Months is distributed across a recommended core team of four specialised roles:

- 1 Senior Backend Engineer (CRM core, Loyalty Engine, SAP integration)
- 1 Database & Integration Engineer (Unified Customer Database, batch sync, ERP bridge)
- 1 Mobile & POS Developer (POS offline agent, Fleet Portal, SMS gateway integration)
- 1 QA & DevOps Lead (test automation, CI/CD pipelines, deployment, UAT coordination)

$$\text{Development Duration} = 12.30 \text{ PM} \div 4 \text{ persons} \approx 3.07 \text{ Core Development Months}$$

Important Note on Timeline Reconciliation: The COCOMO-II model produces a core development effort estimate. The 3.07-month figure represents the active coding and integration phase for the four-person core team. The full project timeline of 12 months (as established in the Assignment 1 business case) accounts for additional phases not captured in the COCOMO-II effort estimate, including requirements elucidation and design (2 months), User Acceptance Testing across 3,500+ sites (3 months), phased regional rollout (3 months), and post-launch stabilisation and hypercare (1 month). The COCOMO estimate and the business case timeline are therefore complementary and mutually consistent.

Step 6: Final Estimation Summary

The following table provides a consolidated reference for all COCOMO-II Early Design estimation parameters and results.

Parameter	Value	Notes
Total UFP	**45 Function Points**	10 requirements across EI, EO, ILF, EIF, EQ
Language Conversion Factor	**50 LOC / FP**	Industry average for Java/C# + C/C++ mixed project
Estimated SLOC	**2,250 SLOC**	45 FP × 50 LOC/FP
KLOC	**2.25 KLOC**	2,250 / 1,000
COCOMO II Constant (A)	**2.94**	Standard COCOMO II Early Design value
Sum of Scale Factors (ΣSF)	**18.57**	PREC+FLEX+RESL+TEAM+PMAT
Exponent B	**1.0957**	0.91 + (0.01 × 18.57)
(KLOC)^B	**2.44**	2.25 ^ 1.0957
Effort Adjustment Factor (EAF)	**1.7157**	Product of all 7 Effort Multipliers
Estimated Effort	**≈ 12.30 Person-Months**	2.94 × 2.44 × 1.7157
Team Size (core)	**4 Persons**	Database Eng., Backend Eng., Mobile Dev., QA Lead

Development Duration	**≈ 3.07 Months**	12.30 PM ÷ 4 persons
Adjusted Timeline (incl. testing/UAT)	**12 Months**	Aligned with business case rollout plan

This COCOMO-II analysis confirms that the PSO CRM project, with a total UFP of 45, a mixed-technology stack, and moderate platform difficulty driven by legacy hardware and low-bandwidth connectivity constraints, is a well-scoped, medium-complexity enterprise project. The estimated 12.30 Person-Months of core development effort is consistent with industry benchmarks for comparable CRM deployments in the energy and utilities sector, and provides PSO's project management office with a defensible, data-driven basis for budgeting, team assembly, and schedule planning.